BABU BANARASI DAS UNIVERSITY

SCHOOL OF COMPUTER APPLICATIONS

ASSIGNMENT ON
DESCRIPTIVE ANALYSIS
(BCADSN23201)

Submitted to:
Ms. Monica Ma'am

Submitted By:
Yash Tandon
BCA(DS & AI)27
1250258513

Contents

1. Note on Data
2. Objectives
3. How This Would Work at Real Scale
4. Top-Selling Products by Quantity
5. Regional Demand — Sales by State
6. Order Status Distribution
7. Category-wise Sales Revenue
8. Top 10 Cities by Sales
9. Quantity Sold by Category
10. Quantity Sold by Price Range
11. Monthly Sales Trend
12. Payment Method Distribution
13. Customer Type Distribution
14. Top 10 Sellers by Sales
15. Average Discount by Product
16. Summary of Key Metrics
17. Next Steps: Descriptive → Predictive Analytics
18. Appendix: Glossary

Descriptive Analytics Report:
Flipkart Product Sales Performance

Domestic Marketplace Sales (January–June 2026)

1. Note on Data

This report uses a Flipkart marketplace sales dataset containing 980 order records across 10 products, 5 product categories, 12 states and 19 cities, covering orders placed between January and June 2026. The dataset includes product, category, unit price, discount, order status, payment method, customer type, seller, and location fields. It is designed to support analysis of product performance, regional demand, category mix, seller performance and customer purchase behaviour.

2. Objectives

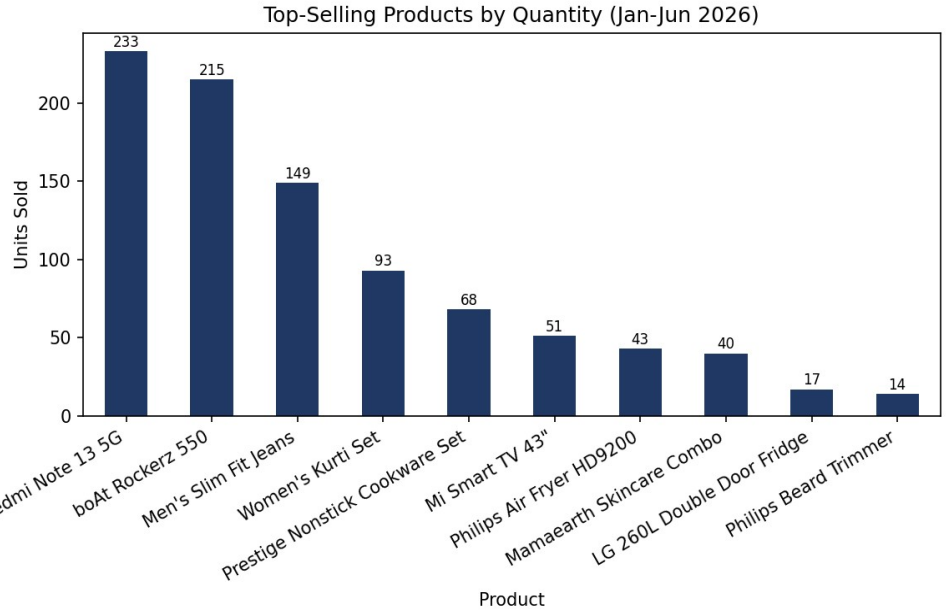
- What are the top-selling products based on quantity sold?
- What is regional sales demand and which states lead in performance?
- What is the distribution of order statuses within the dataset?
- What is category-wise sales revenue and which categories generate the most revenue?
- What are the top 10 cities by sales revenue?
- What is the quantity sold across product categories, and what does this reveal about demand?
- How does quantity sold vary by product price range?
- How does sales performance trend across the six-month period?
- What is the contribution of different payment methods to total sales?
- What is the customer-type mix (first-time buyers vs. repeat buyers vs. Flipkart Plus members)?
- Which sellers are the strongest revenue contributors?
- How does average discounting vary by product?

3. How This Would Work at Real Scale

The table below outlines how this type of descriptive analysis would extend to a production environment operating across Flipkart's full marketplace network.

LAYER	PURPOSE	TYPICAL TOOLS
Data Collection	Store order-level records: product, category, price, discount, location, payment, customer type	Seller Dashboard / Order-Management APIs
Data Loading	Import and load transaction data into the analysis environment	Pandas
Data Cleaning	Check missing values, duplicate order IDs, inconsistent status labels, and standardise state/city names	Pandas
Data Analysis	Compute revenue, unit sales, regional performance, category mix, discount and payment patterns	Pandas, NumPy
Data Visualization	Create charts to identify trends, concentration, and demand patterns	Matplotlib / seaborn
Reporting	Present findings and business insights in a structured report	Word / Jupyter

4. Top-Selling Products by Quantity



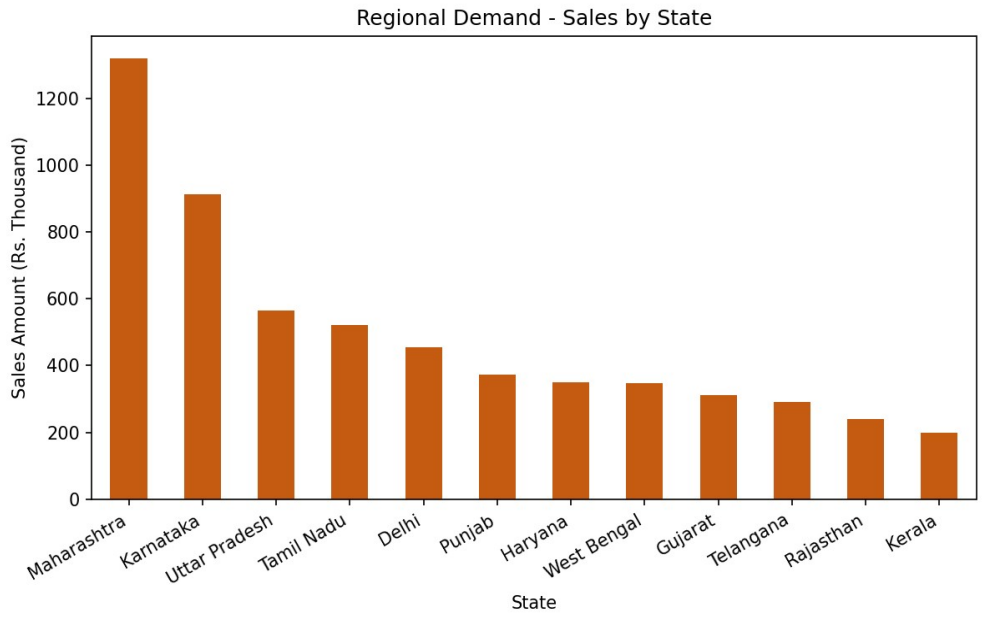
The Top-Selling Products graph ranks Flipkart's product line-up by recorded unit sales across the six-month window. Redmi Note 13 5G is the clear volume leader, with 233 units sold, representing approximately 25.2% of the 923 total units recorded. boAt Rockerz 550 follows closely with 215 units (23.3%), meaning the two most affordable electronics items together account for nearly half of all units sold.

Men's Slim Fit Jeans ranks third with 149 units, ahead of Women's Kurti Set at 93 units. Prestige Nonstick Cookware Set and Mi Smart TV 43" follow at 68 and 51 units respectively, with Philips Air Fryer HD9200 and Mamaearth Skincare Combo close behind at 43 and 40 units. LG 260L Double Door Fridge and Philips Beard Trimmer sit at the bottom of the volume ranking, at 17 and 14 units — expected, given their positioning as high-ticket or niche products.

The pattern is consistent with how a marketplace catalogue typically performs: low-priced, high-frequency electronics and fashion items drive footfall and volume, while large appliances contribute disproportionately to revenue rather than unit count. This distinction becomes clearer in the category revenue analysis in Section 7.

KPI — Top-Selling Products
Top product: Redmi Note 13 5G — 233 units (≈25.2% of total)
Second: boAt Rockerz 550 — 215 units (≈23.3%)
Combined top-2 product share: ≈48.5% of total volume

5. Regional Demand — Sales by State



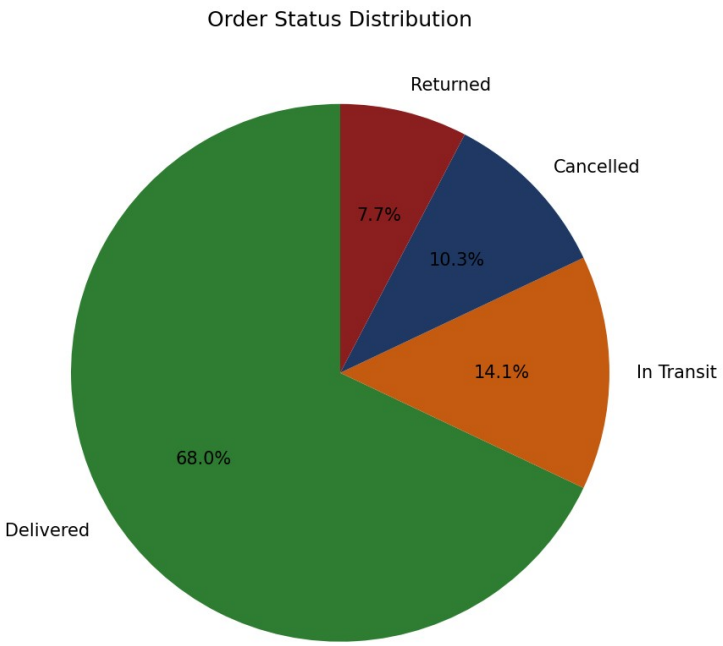
The Regional Demand graph shows that Maharashtra is the leading state by recorded sales, at approximately ₹13,21,062, well ahead of Karnataka at approximately ₹9,13,267. Together these two states account for roughly 37.9% of total recorded revenue, underlining how concentrated Flipkart's demand in this dataset is in India's two largest metro-anchored consumer markets.

Uttar Pradesh ranks third at approximately ₹5,66,012, followed by Tamil Nadu (₹5,22,057) and Delhi (₹4,54,938) in fourth and fifth place. Punjab, Haryana, West Bengal and Gujarat each contribute between ₹3.1 lakh and ₹3.8 lakh, while Telangana, Rajasthan and Kerala form the lower tier of the ranking, with Kerala recording the least regional demand at approximately ₹1,99,730.

The top three states — Maharashtra, Karnataka and Uttar Pradesh — together account for approximately 47.5% of total recorded sales. This concentration matters operationally: it is a reasonable basis for prioritising warehouse stocking, last-mile delivery capacity, and marketing spend toward these markets before investing further in lower-demand states.

KPI — Regional Demand
Top state: Maharashtra — ₹13,21,062 recorded sales
Runner-up: Karnataka — ₹9,13,267
Top-3 state share of total sales: ≈47.5%

6. Order Status Distribution



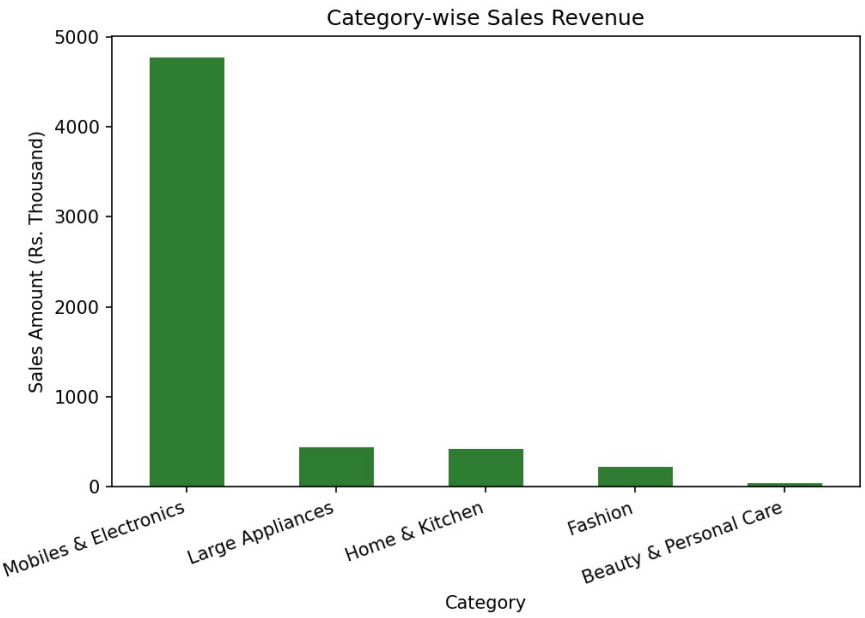
The Order Status Distribution shows that Delivered is the dominant status, accounting for 666 of 980 orders (68.0%). In Transit orders make up 14.1% (138 orders), representing products that have been booked and are moving through the seller warehouse, courier network, or last-mile delivery.

Cancelled orders stand at 10.3% (101 orders) — a figure worth flagging directly rather than glossing over, and Returned orders add a further 7.7% (75 orders). A combined cancellation-and-return rate of roughly 18% on marketplace bookings is high enough to warrant investigation into payment failures, delivery-timeline dissatisfaction, product-quality mismatches, or competitive cross-shopping, rather than being treated as background noise.

When Delivered orders are broken down regionally, Maharashtra leads with 127 deliveries (19.1% of all delivered orders), followed by Karnataka with 115 (17.3%) and Uttar Pradesh with 77 (11.6%). This delivered-order concentration broadly tracks the revenue concentration seen in Section 5.

KPI — Order Status
Delivered: 666 orders (68.0%)
In Transit: 138 orders (14.1%) | Cancelled: 101 orders (10.3%) | Returned: 75 orders (7.7%)
Highest delivered-order state: Maharashtra — 127 orders (19.1%)

7. Category-wise Sales Revenue



The Category-wise Sales analysis evaluates revenue generated by each of Flipkart's five product categories in this dataset: Mobiles & Electronics, Fashion, Home & Kitchen, Beauty & Personal Care, and Large Appliances.

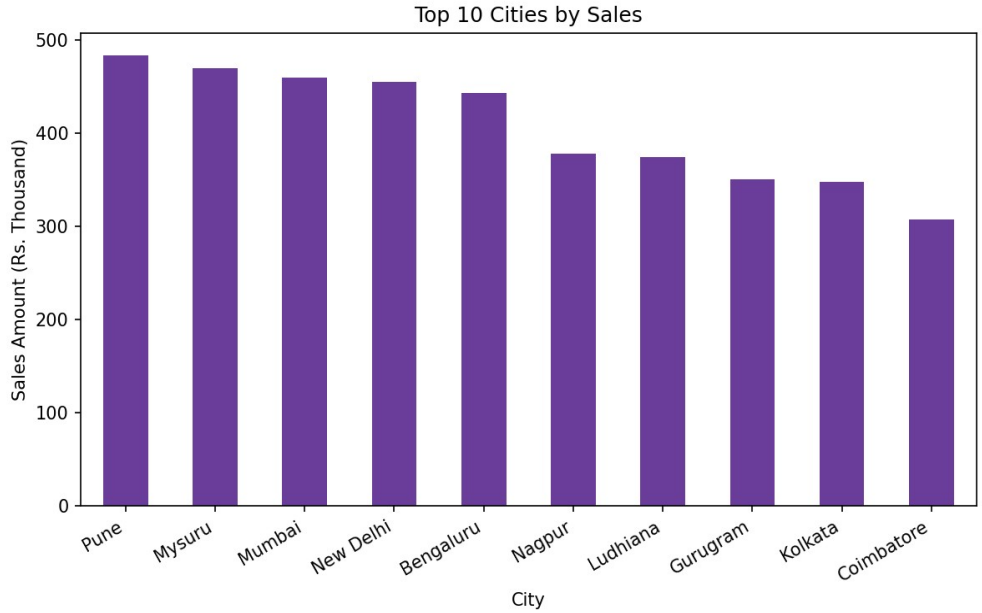
Mobiles & Electronics is by far the highest revenue-generating category, at approximately ₹47,69,239, contributing approximately 80.9% of total recorded sales. This is driven by a combination of the volume-leading Redmi Note 13 5G and boAt Rockerz 550, together with the higher-priced Mi Smart TV 43". Large Appliances ranks second at approximately ₹4,41,127 (7.5%), despite selling only 17 units, reflecting the high unit price of the LG 260L Double Door Fridge.

Home & Kitchen contributes approximately ₹4,18,130 (7.1%), Fashion approximately ₹2,23,427 (3.8%), and Beauty & Personal Care the smallest share at approximately ₹39,746 (0.7%) — unsurprising given the low price point of items in that category.

The gap between Mobiles & Electronics and every other category is much larger in revenue terms than in unit-volume terms, because a small number of high-priced electronics items pull the category's average selling price well above the rest of the catalogue.

KPI — Category Revenue
Highest revenue category: Mobiles & Electronics — ₹47,69,239 (≈80.9%)
Second: Large Appliances — ₹4,41,127 (≈7.5%)
Lowest: Beauty & Personal Care — ₹39,746 (≈0.7%)

8. Top 10 Cities by Sales



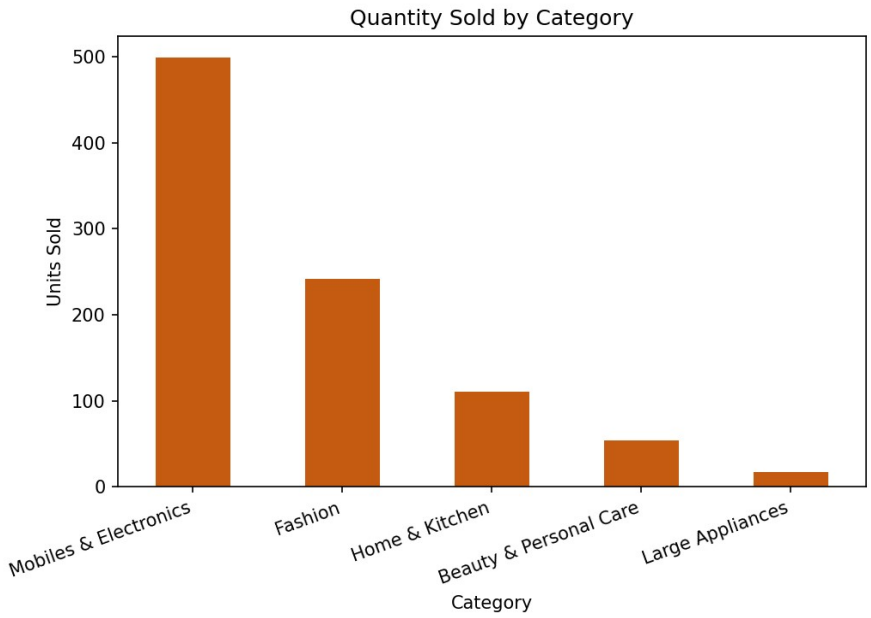
The Top 10 Cities analysis examines sales concentration at the city level. Pune is the leading city, generating approximately ₹4,83,464, or roughly 8.2% of total recorded sales, narrowly ahead of Mysuru (₹4,69,606, 8.0%) and Mumbai (₹4,59,698, 7.8%).

New Delhi follows at approximately ₹4,54,938 (7.7%), with Bengaluru fifth at approximately ₹4,43,661 (7.5%). Nagpur, Ludhiana, Gurugram, Kolkata and Coimbatore form a closely matched next tier, each contributing between ₹3.1 lakh and ₹3.8 lakh.

The four leading cities — Pune, Mysuru, Mumbai and New Delhi — together account for approximately 31.7% of total recorded sales, reinforcing that demand on the platform, while broad-based, still shows a moderate concentration in a set of metro and tier-2 markets with strong internet penetration and established delivery infrastructure.

KPI — Top Cities
Leading city: Pune — ₹4,83,464 (≈8.2% of total sales)
Top 4 cities' combined share: ≈31.7%
Notable tier-2 presence: Mysuru ranks 2nd, Ludhiana ranks 7th overall

9. Quantity Sold by Category



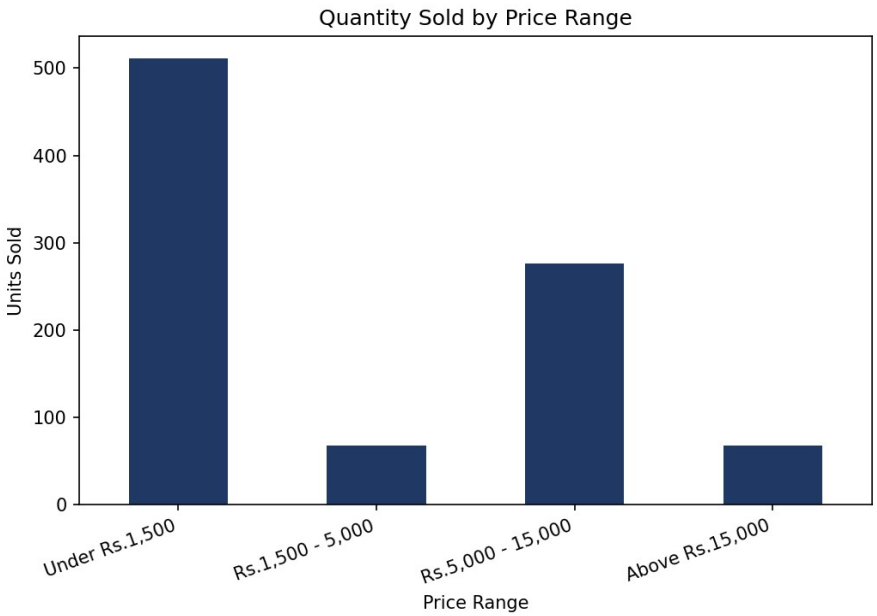
The Quantity Sold by Category analysis measures unit volume rather than revenue. Mobiles & Electronics still leads by volume at 499 units (54.1% of total quantity), though its dominance is less extreme here than in the revenue view, since Fashion sells in high volume at a much lower average price.

Fashion follows at 242 units (26.2%), then Home & Kitchen at 111 units (12.0%), Beauty & Personal Care at 54 units (5.9%), and Large Appliances at just 17 units (1.8%).

Reading Sections 7 and 9 together is the more useful exercise than reading either alone: Mobiles & Electronics dominates both volume and revenue, but its revenue share (80.9%) is disproportionately higher than its volume share (54.1%) because the category's mix includes higher-priced items. Large Appliances shows the opposite pattern — negligible volume share but a meaningful revenue share — which is the kind of category-value asymmetry that should inform inventory and warehouse-space decisions.

KPI — Category Volume
Highest-volume category: Mobiles & Electronics — 499 units (≈54.1%)
Second: Fashion — 242 units (≈26.2%)
Lowest: Large Appliances — 17 units (≈1.8%)

10. Quantity Sold by Price Range



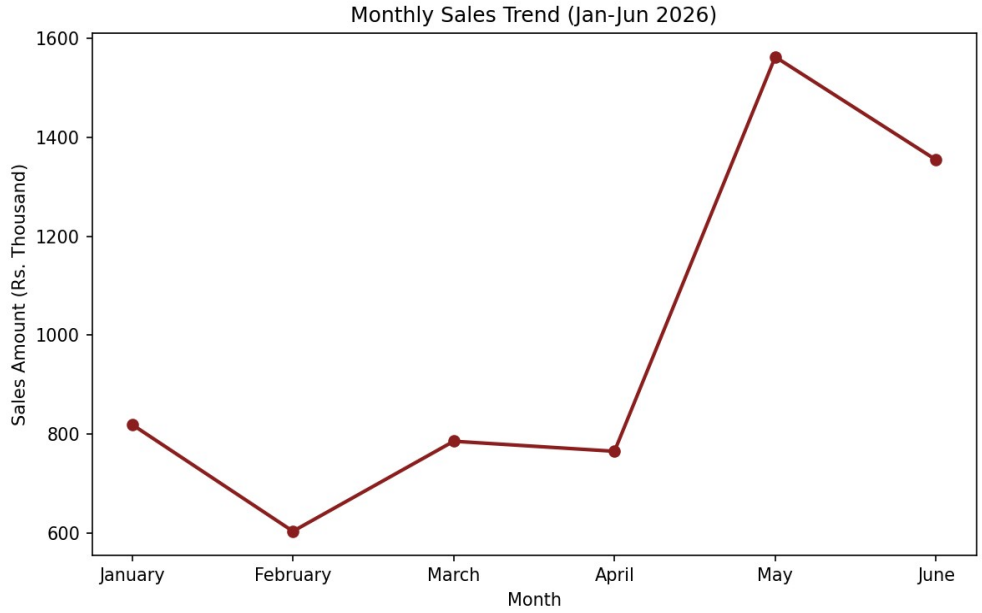
The Quantity Sold by Price Range analysis groups all ten products into four bands: under ₹1,500, ₹1,500–5,000, ₹5,000–15,000, and above ₹15,000.

The under-₹1,500 band — comprising boAt Rockerz 550, Women's Kurti Set, Men's Slim Fit Jeans, Mamaearth Skincare Combo and Philips Beard Trimmer — dominates with 511 units, approximately 55.4% of total recorded quantity. This confirms what Section 4 already suggested: low-ticket, high-frequency items are doing the majority of the volume lifting for the platform in this dataset.

The ₹5,000–15,000 band records 276 units (29.9%), driven largely by Redmi Note 13 5G, while the ₹1,500–5,000 band and the above-₹15,000 band each record 68 units (7.4%). This step pattern is expected in an online marketplace where affordability and frequent replenishment needs push volume toward the lower price tiers, even as higher-priced electronics anchor overall revenue.

KPI — Price Range
Highest-volume band: Under ₹1,500 — 511 units (≈55.4%)
Second: ₹5,000–15,000 — 276 units (≈29.9%)
Lowest (tied): ₹1,500–5,000 and Above ₹15,000 — 68 units each (≈7.4%)

11. Monthly Sales Trend



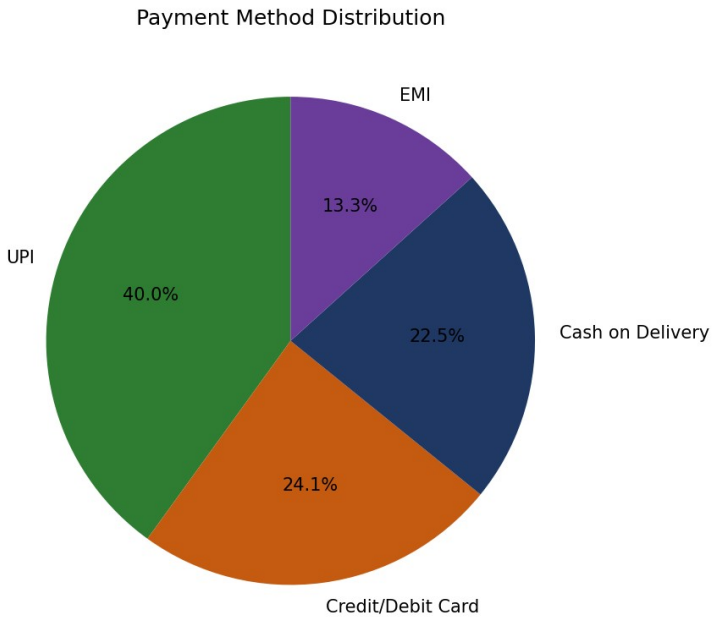
The Monthly Sales Trend tracks recorded revenue from January through June 2026. Sales opened at approximately ₹8,19,435 in January, dipped to ₹6,03,796 in February, then partially recovered through March (₹7,85,777) and April (₹7,65,488).

Sales then rose sharply into May, reaching approximately ₹15,62,664 — more than double the April figure — before settling at approximately ₹13,54,509 in June. This May–June surge is consistent with the pre-summer and early-summer appliance and electronics buying season in India, when demand for items such as air fryers, TVs, and refrigerators typically climbs, and is a plausible driver rather than a random spike.

The practical read here is that May functioned as a genuine seasonal demand step-up rather than a one-off spike, since June held on to most of the gain — a distinction that matters if this data were used to set inventory and staffing targets for the second half of the year. Treating February's dip as the new baseline would understate expected demand for the months ahead, which is exactly the kind of forecasting error Section 17 addresses.

KPI — Monthly Trend
Peak month: May 2026 — ₹15,62,664
Lowest month: February 2026 — ₹6,03,796
April→May growth: ≈104% month-on-month

12. Payment Method Distribution

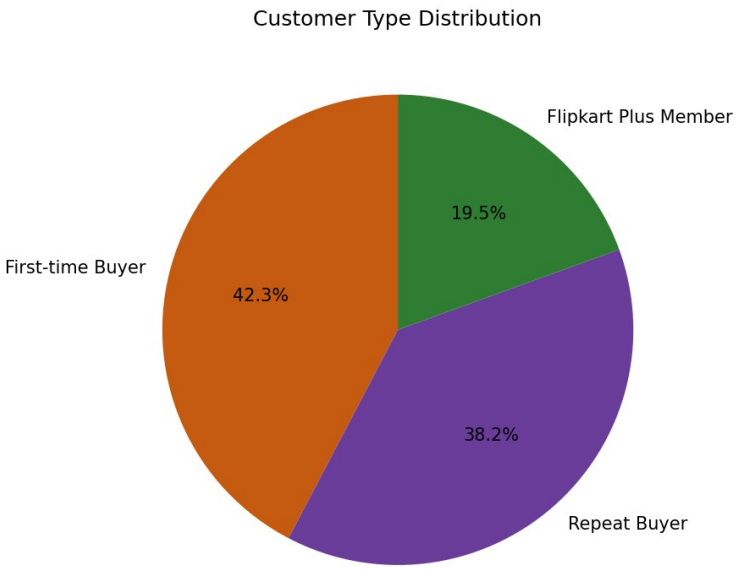


The Payment Method Distribution shows that UPI is the most common payment route, used in 40.0% of transactions (352 orders) — consistent with UPI's position as India's dominant digital-payment rail for online shopping.

Credit/Debit Card accounts for 24.1% (212 orders), Cash on Delivery for 22.5% (198 orders), and EMI for the remaining 13.3% (117 orders). The EMI share is concentrated among higher-priced items such as the Mi Smart TV and LG refrigerator, suggesting financing availability is a meaningful, and possibly under-examined, lever on conversion for big-ticket electronics and appliances.

KPI — Payment Method
Most common: UPI — 40.0% (352 orders)
Credit/Debit Card: 24.1% (212 orders) | Cash on Delivery: 22.5% (198 orders)
EMI: 13.3% (117 orders)

13. Customer Type Distribution

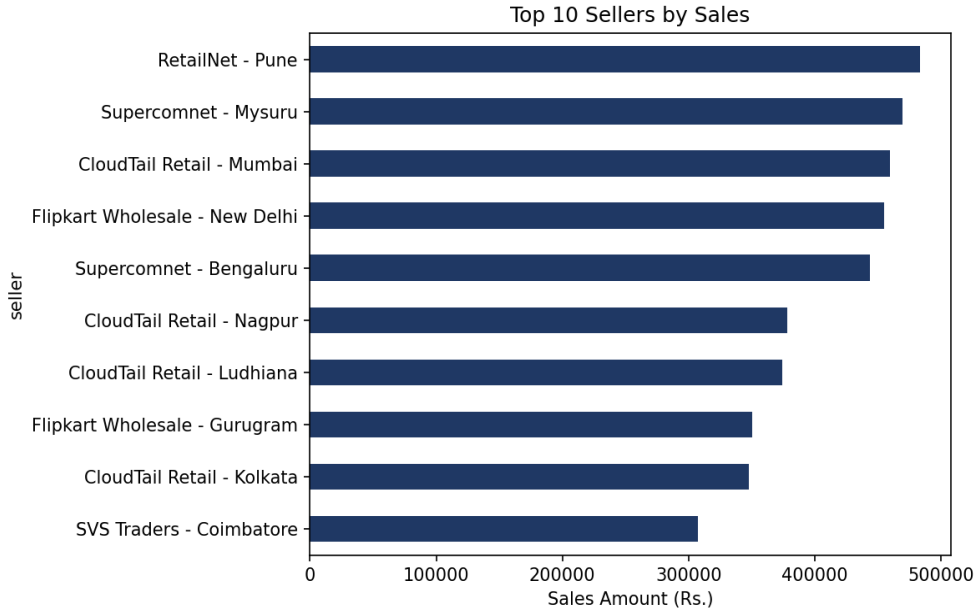


The Customer Type Distribution splits orders into First-time Buyer, Repeat Buyer, and Flipkart Plus Member categories. First-time Buyers form the largest group at 42.3% (372 orders), narrowly ahead of Repeat Buyers at 38.2% (336 orders).

Flipkart Plus Members — the platform's loyalty-programme customers — account for 19.5% (171 orders). This is a healthy sign for customer acquisition, since new buyers make up the largest single group, but the near-equal share of repeat buyers indicates a meaningful base of returning customers as well. Growing the Flipkart Plus share further would be a natural lever for increasing order frequency and average customer lifetime value.

KPI — Customer Type
Largest group: First-time Buyer — 42.3% (372 orders)
Repeat Buyer: 38.2% (336 orders)
Flipkart Plus Member: 19.5% (171 orders)

14. Top 10 Sellers by Sales

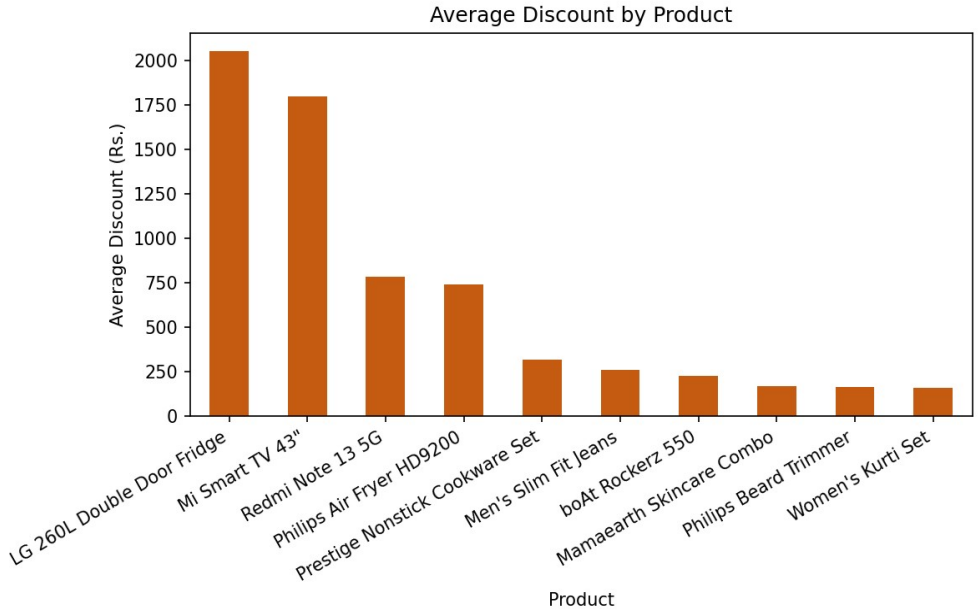


The Top 10 Sellers analysis ranks individual marketplace seller accounts by recorded sales revenue. RetailNet – Pune leads the network at approximately ₹4,83,464, followed by Supercomnet – Mysuru at approximately ₹4,69,606 and CloudTail Retail – Mumbai at approximately ₹4,59,698.

Flipkart Wholesale – New Delhi, Supercomnet – Bengaluru, and a closely bunched group of CloudTail Retail accounts (Nagpur, Ludhiana, Kolkata) round out the top ranking, each contributing between ₹3.1 lakh and ₹4.5 lakh. As expected, this seller ranking closely mirrors the city-level ranking from Section 8, since each city in this dataset maps to a single primary seller account — a limitation of the current data structure that a production system would need to resolve by tracking multiple sellers per city separately.

KPI — Top Sellers
Top seller: RetailNet – Pune — ₹4,83,464
Second: Supercomnet – Mysuru — ₹4,69,606
Third: CloudTail Retail – Mumbai — ₹4,59,698

15. Average Discount by Product



The Average Discount by Product analysis looks at how much price flexibility sellers extend across the catalogue. LG 260L Double Door Fridge carries the highest average discount at ₹2,050 per unit, followed by Mi Smart TV 43" (≈₹1,796) and Redmi Note 13 5G (≈₹782).

Philips Air Fryer HD9200 and Prestige Nonstick Cookware Set sit in the middle of the range at roughly ₹315–₹739, while Women's Kurti Set shows the lowest average discount at ₹160. Read alongside Section 4, this suggests sellers are discounting most aggressively on the highest-priced items — where absolute discount amounts move the needle on conversion — rather than applying uniform percentage discounts across the catalogue.

KPI — Average Discount
Highest average discount: LG 260L Double Door Fridge — ₹2,050/unit
Lowest average discount: Women's Kurti Set — ₹160/unit
Overall average discount across all products: ≈₹512/unit

16. Summary of Key Metrics

METRIC	VALUE
Total order records	980
Usable (non-cancelled) records	879
Total recorded quantity	923 units
Total recorded sales	₹58,91,669
Average order value (AOV)	₹6,703
Top-selling product	Redmi Note 13 5G — 233 units
Top state by sales	Maharashtra — ₹13,21,062
Top city by sales	Pune — ₹4,83,464
Top revenue category	Mobiles & Electronics — ₹47,69,239 (≈80.9%)
Top volume category	Mobiles & Electronics — 499 units (≈54.1%)
Most common order status	Delivered — 666 records (68.0%)
Cancellation rate	10.3% (101 records)
Return rate	7.7% (75 records)
Highest-volume price band	Under ₹1,500 — 511 units (≈55.4%)
Peak sales month	May 2026 — ₹15,62,664
Dominant payment method	UPI — 40.0%
Dominant customer type	First-time Buyer — 42.3%
Top seller	RetailNet – Pune — ₹4,83,464
States represented	12
Cities represented	19
Products represented	10

Read together, these numbers point to one central fact worth stating plainly rather than burying: Flipkart's current volume engine in this dataset is low-ticket electronics and fashion, but its revenue engine is a small number of high-priced electronics and appliance items. Any inventory, marketing or seller-incentive decision built on only one of those two metrics will misread the business.

17. Next Steps: Descriptive → Predictive Analytics

The analysis above describes what has already happened. The more useful — and harder — next step is estimating what is likely to happen next, and where the current descriptive picture could actively mislead a decision-maker if treated as a forecast.

17.1 Demand Forecasting by Product and Region

Historical order data — product, category, state, city, month, and payment method — can be used to build month-ahead demand estimates per product per state, rather than relying on trailing three-month averages. This matters directly here: naively projecting May's peak forward would overstate H2 2026 demand, as Section 11 already flags.

17.2 Inventory and Fulfilment Planning

Redmi Note 13 5G and boAt Rockerz 550 drive nearly half of unit volume; a stock-out on either product at a top warehouse (Pune, Mysuru) carries disproportionate revenue risk. A predictive reorder-point model, weighted by each fulfilment centre's historical sell-through rate, would reduce both stock-out and overstock exposure — currently this report can only describe past concentration, not recommend a specific reorder threshold.

17.3 Cancellation and Return Risk Modelling

A combined cancellation-and-return rate of roughly 18% is high enough to model rather than monitor. A logistic model using payment method, discount level, customer type, and time-in-transit as predictors could flag at-risk orders early enough for a seller to intervene — a Cash-on-Delivery order sitting In Transit for an unusually long period is a plausible leading indicator worth testing against this dataset structure.

17.4 Regional and Seller Expansion Scoring

Kerala and Rajasthan currently show the lowest regional demand. Before concluding these are low-priority markets, a predictive model incorporating population, internet penetration, and existing e-commerce delivery-network density would separate genuinely low-potential markets from under-served ones with thin fulfilment coverage — the current data cannot distinguish between the two.

17.5 Discount Optimisation

The discount pattern in Section 15 is descriptive, not causal. A predictive elasticity model — estimating how conversion probability changes with discount depth, by product and customer type — would let sellers move from a flat, catalogue-wide discount approach to one calibrated against actual price sensitivity.

18. Appendix: Glossary

Order Status — The current stage of an order: Delivered, In Transit, Cancelled, or Returned.

Sales Amount — The monetary value of a transaction after discount, used to calculate total and category-wise sales.

Category — The product category a listing belongs to: Mobiles & Electronics, Fashion, Home & Kitchen, Beauty & Personal Care, or Large Appliances.

Price Range — A grouping of products by unit price, used as a proxy for pricing tier and target buyer.

Regional Demand — The level of sales or quantity recorded across states and cities.

AOV — Average Order Value; total sales divided by the number of usable (non-cancelled) orders.

Flipkart Plus — Flipkart's loyalty membership programme, offering benefits such as early access to sales and faster delivery.

KPI — Key Performance Indicator; a measurable value used to evaluate an important aspect of sales performance.

Descriptive Analysis — The process of summarising historical data using KPIs, tables and charts to understand existing sales patterns, as distinct from predictive analysis, which estimates future outcomes.